

GAWTAM CHITHRA RAMESH

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Summary

Robotics engineer (MSc, KTH) with hands-on experience deploying industrial robots and cobots in production, from real-time controllers and trajectory generation to ROS-based motion planning. Combines factory automation with computer vision and adaptive AI to improve safety, quality, and productivity, balancing near-term delivery with exploration of emerging physical-AI technology.

Education

KTH Royal Institute of Technology

Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0)

Aug 2024 – Jun 2026

Stockholm, Sweden

Indian Institute of Technology Madras

Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10)

Aug 2019 – Jul 2024

Chennai, India

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, ROS/ROS2, MoveIt, Gazebo, MuJoCo, PyTorch, OpenCV, Behavior Trees, Docker, CUDA, MATLAB

Professional Experience

Master's Thesis: Generative Planning for Industrial Manipulation

ABB Robotics

Jan 2026 – Jun 2026

Västerås, Sweden

- Designed a generative flow-matching motion planner for industrial manipulation steered by formal task goals.
- Steered the pretrained planner to new task constraints at inference without retraining for fast redeployment.

Device Research Intern: 3D Scene Graphs

Ericsson Research

Jul 2025 – Dec 2025

Lund, Sweden

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph perception framework.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.
- Benchmarked state-of-the-art vision models to balance reconstruction fidelity against real-time inference latency.

Graduate Robotics Intern: Collaborative Robots

Systemantics India

Dec 2022 – Jul 2023

Bangalore, India

- Built a complete real-time system controller from scratch for a 6-DoF collaborative robot on Linux PREEMPT-RT.
- Generated jerk-limited S-curve trajectories that cut mechanical shock via smooth acceleration and deceleration.
- Engineered behavior-tree task planning and ROS2 MoveIt motion planning for long-horizon pipetting and inspection.

Robotics Intern: Pallet Handling

Yaskawa

Jun 2021 – Jul 2021

Gurgaon, India

- Modeled 3D palletizing layouts for mixed-size cartons on an industrial robot using offline simulation tooling.
- Programmed pick-and-place sequences on the teach pendant and validated reachability across varied carton stacks.
- Wrote a C++ routine to optimize controller memory while streaming motion data from a PC to the robot controller.

Research Experience

Autonomous Ultrasound Scanning from a Single RGB Camera

Nirav Patel, INSPIRE Lab, IIT Madras

Jan 2024 – May 2024

Chennai, India

- Reconstructed a dense 3D torso point cloud from one RGB camera via structure-from-motion and multi-view stereo.
- Segmented the abdominal region from the point cloud and estimated surface tangents for the probe approach angle.
- Closed a perception-to-planning loop with ROS and MoveIt motion planning validated in physics simulation.

Technical Projects

Robot Learning and Embodied AI | DD2600, KTH

Aug 2025 – Oct 2025

- Built a vision-language pipeline mapping RGB-D frames to 3D semantic voxel maps for queryable object presence.
- Trained a ResNet classifier on 2,000 heatmap samples, reaching 86.2% validation accuracy and 0.917 ROC-AUC.

Image Analysis and Computer Vision | DD2423, KTH

Oct 2024 – Jan 2025

- Implemented edge detection via differential-geometry zero-crossings and a Hough-transform line detector.
- Built SIFT feature matching with RANSAC homography estimation for robust planar-scene alignment.
- Estimated the fundamental matrix and triangulated 3D points from calibrated camera projection matrices.

Introduction to Robotics | DD2410, KTH

Sep 2024 – Oct 2024

- Implemented analytic and Jacobian-based inverse kinematics for 3-DoF and 7-DoF manipulators, passing all tests.
- Built a 2D occupancy-grid mapper from laser scans with raytraced free space and configuration-space inflation.
- Designed a behavior-tree mission planner for a mobile manipulator performing vision-based grasping in simulation.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26**).(Under Review)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25 Workshop**)